SOWETO GLOBAL SOLUTIONS

IT KNOWLEDGE SERIES & PRACTICAL GUIDES

SERVER & NETWORK ADMINISTRATION

A Step-by-Step Practical Guide to LANs, Wi-Fi, Backups, and User Access

Designed for students, business owners, and aspiring IT administrators. This educational guide breaks down fundamental networking concepts into actionable, step-by-step setup instructions to build and secure your own network infrastructure.

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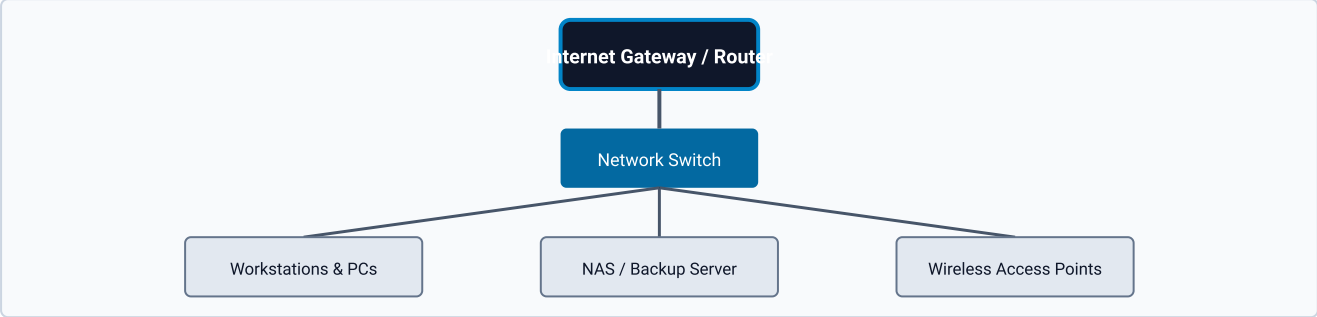
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1.0 LOCAL AREA NETWORK (LAN) SETUP

A **Local Area Network (LAN)** connects computers and devices within a limited area such as an office, home, or building. Setting up a LAN correctly ensures high data transfer speeds, minimal connectivity drops, and strong security.

> **Key Concept:** A standard LAN setup uses an Internet Router, a Network Switch (to add more wired ports), and Ethernet cables (Cat6 or Cat6a) connected to end devices like PCs, servers, and smart devices.

1.1 Understanding Network Topology



1.2 How to Set Up Your LAN (Step-by-Step)

STEP 1 **Connect Hardware:** Plug an Ethernet cable from your ISP modem/fiber box into the WAN port of your router. Run another Ethernet cable from a router LAN port into your network switch.

STEP 2 **Access Router Dashboard:** Open a web browser on a connected PC and type your router's default IP address (usually `192.168.1.1` or `192.168.0.1`). Log in using admin credentials.

STEP 3 **Configure DHCP Server:** Enable DHCP (Dynamic Host Configuration Protocol) so your router automatically hands out IP addresses to connected devices (e.g., range `192.168.1.100` to `192.168.1.200`).

STEP 4 **Assign Static IP Addresses:** Assign permanent, fixed IP addresses to servers, printers, and backup drives so they never change location on your network.

2.0 WI-FI CONFIGURATION & SECURITY

A properly configured wireless network provides convenience while shielding your network from unauthorized intruders. Learning how to properly name and secure your Wi-Fi is essential for any modern network.

2.1 Wi-Fi Setup Quick Reference

Setting Category	Recommended Setting	Why It Matters
Security Protocol	WPA3-Personal or WPA2/WPA3 Mixed	Protects traffic with modern, uncrackable encryption algorithms.
Frequency Band	Dual-Band (2.4 GHz + 5 GHz)	2.4 GHz offers long range; 5 GHz offers high-speed transfer.
Guest Network	Enabled (Isolated)	Allows visitors internet access without seeing your private files.

2.2 Step-by-Step Wi-Fi Setup Guide

STEP 1 **Set Network Name (SSID):** Choose a unique name for your wireless network. Avoid using personal names or address details in your SSID.

STEP 2 **Set Security Mode & Password:** Select **WPA3-Personal** (or WPA2-PSK AES). Create a passphrase that is at least 12–16 characters long using a mix of letters, numbers, and symbols.

STEP 3 **Enable Guest Network Isolation:** Turn on your router's "Guest Wi-Fi" feature. Ensure "Client Isolation" is checked so guests cannot access your local servers or devices.

3.0 SERVER DATA BACKUP & RECOVERY

Hard drives fail, files get accidentally deleted, and cyber threats like ransomware can lock data. A reliable server backup system guarantees you can recover your files at any time.

> **The Golden 3-2-1 Rule:** Keep **3** copies of important files, across **2** different media types (e.g., local server + external hard drive), with **1** copy kept offsite in the cloud.

3.1 Backup Strategy Overview

Backup Type	How It Works	Best Use Case
Full Backup	Copies every single file and folder.	Run once a week (e.g., Sunday night).
Incremental Backup	Only copies files changed since the last backup.	Run daily or hourly to save storage space.
Cloud Backup	Uploads encrypted data to remote cloud storage.	Protects against fire, theft, or local equipment damage.

4.0 USER ACCESS MANAGEMENT (UAM)

User Access Management regulates who can log into your computers, server drives, and software applications. Managing permissions prevents accidental file deletion and blocks unauthorized users.

4.1 Role-Based Access Control (RBAC) Matrix

User Account Level	System Privileges	Recommended Best Practice
Standard User	Can run applications and edit personal files. Cannot install software or change network settings.	Assign this account type to all day-to-day users to prevent malware infections.
Administrator	Full control over hardware, operating systems, user creation, and security options.	Use only when performing system updates or installing new software.
Guest / Limited	Temporary internet access only. No file storage rights.	Disable by default when not actively in use.

4.2 Step-by-Step User Management Workflow



STEP 1 **Create Individual Accounts:** Never share a single account among multiple people. Create unique usernames for each person so actions can be audited.

STEP 2 **Enforce Strong Passwords:** Require passwords to be at least 12 characters long, including uppercase letters, numbers, and symbols. Encourage the use of a password manager.

STEP 3 **Enable Multi-Factor Authentication (MFA):** Turn on MFA / 2FA across all email and server logins using authenticator apps (e.g., Google Authenticator, Microsoft Authenticator).

Summary Checklist for Administrators

Keep this quick checklist handy when setting up or auditing any network environment:

- [ ] Router default administrator password changed from default.
- [ ] WPA3 or WPA2-AES encryption active on all Wi-Fi networks.
- [ ] Daily automated backups configured with at least one offsite backup.
- [ ] Standard user accounts used for daily operations instead of Admin accounts.